Experiment No. 1

Aim: Implement programs to demonstrate control flow and arrays using Java.

1. Implement a Java program to find the factorial of a given number.

```
import java.util.Scanner; public class
FactorialProgram {     public static void
main(String[] args) {
       Scanner sc = new Scanner(System.in);
System.out.print("Enter a number: ");
int num = sc.nextInt();         long factorial =
1;         if (num < 0) {
           System.out.println("Factorial not defined for negative numbers.");
       } else {            for (int i = 1; i
<= num; i++) {                factorial =
factorial * i;
           }
           System.out.println("Factorial of " + num + " is: " + factorial);
       }
sc.close();
   }
}
```

```
PS D:\Prasanna\exp1> javac FactorialProgram.java
PS D:\Prasanna\exp1> java FactorialProgram
Enter a number: 5
Factorial of 5 is: 120
PS D:\Prasanna\exp1>
```

2. Implement a Java program to check whether a given number is prime or not. (Take the number as a command-line argument.)

```
public class PrimeCheck {     public
static void main(String[] args) {         if
(args.length == 0) {
            System.out.println("Please provide a number as a command-line argument.");
            return;
        }
        int num = Integer.parseInt(args[0]);
boolean isPrime = true;         if (num <=
1) {            isPrime = false;
        } else {            for (int i = 2; i <=
Math.sqrt(num); i++) {                if (num % i ==
0) {                    isPrime = false;
break;
                }
            }        }        if (isPrime) {
            System.out.println(num + " is a
Prime Number.");
        } else {
            System.out.println(num + " is NOT a Prime Number.");
        }
    }
}
```

```
PS D:\Prasanna\exp1> javac Prime.java
PS D:\Prasanna\exp1> java Prime
Enter a number: 73
Prime Number
PS D:\Prasanna\exp1> javac Prime.java
PS D:\Prasanna\exp1> java Prime
Enter a number: 100
Not Prime Number
PS D:\Prasanna\exp1>
```

3. Implement a Java program to sort a given list of 10 numbers in ascending order.

```
import java.util.Scanner; public class
SortTenNumbers {     public static void
main(String[] args) {         Scanner sc =
new Scanner(System.in);         int[] arr =
new int[10];
        System.out.println("Enter 10 numbers:");
for (int i = 0; i < 10; i++) {             arr[i] =
sc.nextInt();
        }         for (int i = 0; i < arr.length - 1; i++)
{             for (int j = 0; j < arr.length - 1 - i; j++)
{                if (arr[j] > arr[j + 1])
{                    int temp = arr[j];
arr[j] = arr[j + 1];                    arr[j +
1] = temp;
                }
            }
        }
        System.out.println("Numbers in
ascending order:");         for (int
num : arr) {
            System.out.print(num + " ");
        }
sc.close();
    }
}
```

```
PS D:\Prasanna\exp1> javac SortNumbers.java
PS D:\Prasanna\exp1> java SortNumbers
Enter 10 numbers:
34
23
12
34
22
14
65
78
24
56
Numbers in ascending order:
12 14 22 23 24 34 34 56 65 78
PS D:\Prasanna\exp1>
```

4. Implement a Java program to merge two sorted arrays.

```
import java.util.Scanner; public class
MergeSortedArrays {     public static
void main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Enter size of first array: ");
int n1 = sc.nextInt();          int[] a = new int[n1];
        System.out.println("Enter elements of first sorted array:");
for (int i = 0; i < n1; i++) {              a[i] = sc.nextInt();
        }
        System.out.print("Enter size of second array: ");
int n2 = sc.nextInt();          int[] b = new int[n2];
        System.out.println("Enter elements of second sorted array:");
for (int i = 0; i < n2; i++) {              b[i] = sc.nextInt();
        }
        int[] merged = new int[n1 + n2];
int i = 0, j = 0, k = 0;

        while (i < n1 && j < n2)
{               if (a[i] <= b[j])
{                   merged[k++] =
a[i++];
            } else
{                   merged[k++] =
b[j++];             }

        }       while (i < n1)
{               merged[k++] =
a[i++];
```

```
        }        while (j < n2)
{            merged[k++] =
b[j++];
        }
        System.out.println("Merged array:");
for (int num : merged) {
            System.out.print(num + " ");
        }
sc.close();
    }
}
```

```
PS D:\Prasanna\exp1> javac MergeSortedArrays.java
PS D:\Prasanna\exp1> java MergeSortedArrays
Enter size of first array: 5
Enter elements of first sorted array:
23
33
44
55
66
Enter size of second array: 5
Enter elements of second sorted array:
34
45
56
67
78
Merged array:
23 33 34 44 45 55 56 66 67 78
PS D:\Prasanna\exp1>
```

5. Implement a Java program to perform 2×2 matrix multiplication, addition, and transpose (using a switch case).

```
import java.util.Scanner; public class
MatrixOperations {     public static void
main(String[] args) {         Scanner sc =
new Scanner(System.in);         int[][] A =
new int[2][2];         int[][] B = new
int[2][2];         int[][] result = new int[2][2];
       System.out.println("Enter elements of Matrix A:");
for (int i = 0; i < 2; i++)            for (int j = 0; j < 2; j++)
              A[i][j] = sc.nextInt();
       System.out.println("Enter elements of Matrix B:");
for (int i = 0; i < 2; i++)            for (int j = 0; j < 2; j++)
B[i][j] = sc.nextInt();         while (true) {             for (int
i = 0; i < 2; i++)                for (int j = 0; j < 2; j++)
result[i][j] = 0;
           System.out.println("1.Addition  2.Multiplication  3.Transpose  4.Exit");
int choice = sc.nextInt();            switch (choice) {                case 1:
                  for (int i = 0; i < 2; i++)
for (int j = 0; j < 2; j++)
result[i][j] = A[i][j] + B[i][j];
System.out.println("Result:");                    for
(int i = 0; i < 2; i++) {                     for (int j
= 0; j < 2; j++)
                        System.out.print(result[i][j] + " ");
                     System.out.println();
                  }
break;
case 2:
```

```
                for (int i = 0; i < 2; i++)
for (int j = 0; j < 2; j++)                    for (int k
= 0; k < 2; k++)                    result[i][j] +=
A[i][k] * B[k][j];
System.out.println("Result:");                for (int i
= 0; i < 2; i++) {                    for (int j = 0; j < 2;
j++)
                    System.out.print(result[i][j] + " ");
                System.out.println();
                }
break;
case 3:
                System.out.println("Transpose of Matrix A:");
for (int i = 0; i < 2; i++) {                    for (int j = 0; j <
2; j++)
                    System.out.print(A[j][i] + " ");
                System.out.println();
                }
break;
case 4:
                sc.close();
                return;
default:
                System.out.println("Invalid choice");
            }
        }
    }
}
```

```
PS D:\Prasanna\exp1> javac MatrixOperations.java
PS D:\Prasanna\exp1> java MatrixOperations
Enter elements of Matrix A:
23
43
54
65
Enter elements of Matrix B:
11
22
33
44
1.Addition  2.Multiplication  3.Transpose  4.Exit
1
Result:
34 65
87 109
1.Addition  2.Multiplication  3.Transpose  4.Exit
2
Result:
1672 2398
2739 4048
1.Addition  2.Multiplication  3.Transpose  4.Exit
3
Transpose of Matrix A:
23 54
43 65
1.Addition  2.Multiplication  3.Transpose  4.Exit
4
PS D:\Prasanna\exp1>
```